K. S. INSTITUTE OF TECHNOLOGY, BENGALURU-560109
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT PHASE - I (BCS685)

PROJECT TITLE: Indian Market Portfolio Intelligence & Backtesting Platform

Guide's Name with Signature: Beena K. Pradeep

Batch No.: 2026_CSE_01

Sl. No.	NAME	USN
1	Aryaman Tiwari	1KS23CS020
2	Chirag T.	1KS23CS034
3	Durgashree M.	1KS23CS043
4	Jayaditya Dev	1KS23CS059

ABSTRACT

Financial markets exhibit regime shifts where the effectiveness of trading strategies varies across different conditions. Many retail investors adopt strategies without evaluating their robustness or risk-adjusted performance. This project proposes a **Portfolio Intelligence and Backtesting Platform** that evaluates investment strategies on Indian equity markets using historical daily data.

Historical OHLC market data is processed to generate features such as returns, volatility, momentum, and drawdown. A structured backtesting engine evaluates predefined strategies - including Buy & Hold, Moving Average Crossover, RSI-based trading, Momentum, and Mean Reversion - using performance metrics such as CAGR, Sharpe Ratio, volatility, and maximum drawdown.

The system consists of three core modules

- **Market Regime Detection Module** – Uses unsupervised learning techniques such as K-Means or Gaussian Mixture Models to identify distinct market conditions based on returns and volatility.
- **Strategy Evaluation & Suitability Module** – Performs backtesting and applies supervised models (e.g., Random Forest or Gradient Boosting) to estimate the probability that a strategy will outperform a benchmark under the current regime.
- **Risk Forecasting Module** – Estimates potential downside exposure using volatility modeling and probabilistic drawdown analysis.

The platform produces outputs including performance metrics, equity curves, regime-wise strategy analysis, ML-based strategy recommendations, and risk forecasts. This approach provides a structured framework for evaluating investment strategies across varying market conditions.

Keywords: Portfolio Backtesting, Financial Time-Series Analysis, Market Regime Detection, Strategy Evaluation, Machine Learning in Finance, Risk Forecasting.

K. S. INSTITUTE OF TECHNOLOGY, BENGALURU-560109

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT PHASE - I (BCS685)

System Requirements (H/W and S/W)

Minimum Hardware Requirements:

- **Processor:** A modern multi-core processor (e.g., Intel Core i5/i7 or AMD Ryzen 5/7).
- **RAM:** 8 GB or higher.
- **Storage:** A SSD with at least 5 GB of space for the project, dataset, and libraries.
- **Internet Connectivity:** Required for downloading historical market data.

Minimum Software Requirements:

- **Operating System:** Windows 10/11, macOS, or a Linux distribution.
- **Programming Language:** Python 3.9 or higher.
- **Libraries:** Pandas, NumPy, scikit-learn, Plotly, yfinance/NSEpy, FastAPI
- **Database:** PostgreSQL
- **Development Environment:** Any standard IDE like VS Code or PyCharm.

Base Paper Submitted: Yes

K. Alam, M. H. Bhuiyan, I. U. Haque, M. F. Monir, and T. Ahmed, *"Enhancing Stock Market Prediction: A Robust LSTM-DNN Model Analysis on 26 Real-Life Datasets,"* IEEE Access, vol. 12, pp. 122757–122768, Jul. 2024.

ACCEPTED	REJECTED	RE SUBMIT
Reason for Rejection:		
Reason for Re Submit:		

Project Coordinators **HoD**

Prof. Roopesh Kumar B N

Prof. Raghavendrachar S